Violent Crime, Indigenous Institutions and Social Capital: Evidence from Mexico

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Abstract

Social capital, particularly interpersonal trust, is vital for economic growth, especially in developing nations. Despite recent work, it is still not currently clear how violence, particularly in the context of crime, impacts trust and social capital. I investigate the question in the context of the increase of violent crime due the Mexican government's decision to militarize actions against drug cartels. Using a panel based survey, I determine the effect on social capital - particularly interpersonal trust - due to the surge in violence. I find that as violent crime increases, there is an associated decrease in trust; this reduction is attenuated in municipalities with strong Indigenous institutions stemming from precolonial Mexican polities. I provide evidence that a fear response may drive the heterogeneous results for social capital.

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1 Introduction

Social Capital

High levels of social capital and interpersonal trust are crucial for economic growth. They reduce transaction costs and encourage innovation; parties that trust each other spend less resources on legal services and guarding against theft (Knack and Keefer, 1997). High levels of trust can also be beneficial when formal institutions are weak. It enables the emergence of informal credit markets in contexts where individuals are sub-optimally credit constrained (Knack and Keefer 1997). In regions with limited government capacity, high levels of trust enables voluntary collection action to provide public goods, as it reduces fears that others may free-ride (Darluf and Fafchamps 2005).

Recent findings have shown that exposure to violence has an impact on social capital. While, some authors have found a positive impact of experiencing or recalling violence on social capital and other prosocial behaviour (Bellows and Miguel 2009; Voors et al. 2012; Bogliacino, Cangrejo, and Grimalda 2023; surveyed in Bauer at al. 2016), other authors (Kijewski and Frietag 2018; Rohner, Thoenig, and Zilibotti 2013; surveyed in Fiedler 2023) find a negative impact of violent conflict on social capital. These mixed results may suggest heterogeneity in the effects on social capital. [1]

Understanding therefore the impact of violence on trust, and social capital more generally, especially investigating sources of heterogeneity, remains an open question. I look at one source of heterogeneity - pre-existing sociocultural institutions in the context of the Mexican Drug War. I use a panel dataset of Mexican households to compare interpersonal trust before and after a quasi-exogenous surge of violent crime. I further examine heterogeneous impacts by looking at how the impact on trust differs by the strength of preexisting institutions within a municipality, specifically precolonial Indigenous institutions. I also look at the mechanism

[1] In fact, part of this heterogeneity may stem from a parochial effect. Bauer et al. (2016) argue that violence may impact one's in-group - violence may have the largest impact towards increasing trust and cooperative behaviour among members of the same community, village, or ethnic group.

that might be driving the impacts on social trust.

Specifically, I use data from the Mexican Family Life Survey (MxFLS) to investigate how interpersonal trust changes in response to violence. The MxFLS is a nationally representative panel of the Mexican population in 2002 - the year the baseline survey was conducted. Two followup surveys were also conducted between 2005-2006 and 2009-2012 respectively (Rubalcava and Teruel 2013b). The longitudinal nature of the dataset therefore allows trust levels to be compared across different waves among the *same* household. Critically, the second survey (2005-2006) took place before a major escalation in violent crime, while the third survey (2009-2012) was conducted after a dramatic increase in violence. This increase in violent crime can be attributed to former President Calderon's decision to launch a 'War on Drugs', militarizing anti-drug enforcement operations. This militarization led to more violence, with crime rates reaching then-record levels by mid 2008 and fighting spreading to large swaths of Mexican society, even in municipalities that had not seen high crime levels before (Guerrero-Gutiérrez 2011). I proxy for crime by taking the homicide rate per 10,000 people (in the past 12 months) for each municipality, and find how intra-community level trust responds as the level of local violence changes over time.

I also investigate heterogeneity based on preexising social ties. In particular, following Elizalde (2020), I construct a measure of precolonial Indigenous institutions using the *Jurisdictional Hierarchy Beyond the Local Community* measure from Murdock's (1967) *Ethnographic Atlas*. Given that these Indigenous institutions have persisted even through Spanish colonization and the founding of the modern Mexican state, I proxy for informal social networks based on the strength of these Indigenous institutions. I then examine whether there is heterogeneity in changes of trust levels in response to violence depending on Indigenous institutional strength within a municipality.

Overall, I find evidence that increasing violence decreases social trust, particularly among Indigenous households; an increase in the homicide rate by 1 per 10,000 people increases the proportion of Indigenous households who (a) strongly disagree that they can trust community

members by 0.97 percentage points and (b) disagree or strongly disagree by 1.73 percentage points (compared to an average of 1.96% and 16.5% before the escalation of violence). This however masks heterogeneity. I find that losses to social trust are attenuated in municipalities with strong Indigenous institutions. In municipalities with weak Indigenous instutions (a value of 0), a unit increase in the homicide rate results in: (a) a 2.88 percentage point increase in the proportion of households who strongly disagree their neighbours can be trusted, and (b) a 5.44 percentage point increase in the proportion who disagree or strongly disagree. For every unit increase in the Indigenous institutions measure, these increases are attenuated by 1.73 pp and 3.46 pp (at the 10% significance level) respectively.

I also investigate what mechanisms may explain these results. There exist many potential mechanisms for how violence may impact social capital. Violence may lead individuals to brand together to protect themselves against an outside threat encouraging neighbourhood unity and leading communities to a more cooperative equilibrium. Alternatively, conflict may lead to 'post traumatic growth', where survivors gain a new perspective on life, and place increased value on relationships with friends and family. A third mechanism can be labelled as 'post traumatic withdrawal', where conflict causes distress, as individuals feel less close to, and withdraw from others (See reviews in Bauer et al. 2016; Fielder 2023). I investigate these mechanisms in this paper, I find results suggesting a fear response is the mechanism driving the declines in community level trust, and is consistent with the explanations around post-traumatic withdrawal.

This paper adds to the literature on the impact of violence on interpersonal trust and social capital. Over the last few decades, a growing literature has studied the impact of wartime conflict on interpersonal trust and social capital (Bellows and Miguel 2009; Voors et al. 2012; Rohner, Thoenig, and Zilibotti 2013; Kijewski and Frietag 2018; Hoch, Pondorfer, and Shkola 2025). Conversely, there has been a paucity of causal evidence looking at the impact of a surge in non-war related violence on interpersonal trust (though see Leonardo Becchetti, Conzo and Romeo 2014; Rubio-Ramos 2024.) I add to this understudied strand

of literature by leveraging a plausibly exogenous surge of drug related conflict in Mexico. This paper also adds to the overall conflict literature by providing additional support for the theory of post-traumatic withdrawal caused by violence (Kijewski and Freitag 2018).

Furthermore, this paper adds to the growing literature documenting the impact of pre-colonial institutions on modern day outcomes. Precolonial institutions in Vietnam have been associated with the persistence of local cooperation norms (Dell, Lane, and Querubin 2018), while areas with more complex and centralized precolonial Indigenous institutions tend to have better modern economic outcomes (Gennaioli and Rainer 2007; Michalopoulos and Papaioannou 2013) and are more successful in defending their interests against colonizers and post-colonial states (Angeles and Elizalde 2017; Elizalde 2020). I contribute to this literature by providing suggestive evidence that areas with more centralized pre-colonial Indigenous institutions are able to mitigate losses in trust due to a surge in violent crime.

Lastly, this paper adds to the literature documenting the socio-economic costs of the ongoing Mexican Drug War. Prior work has found that the drug war has lead to a rise in risk aversion (Brown et al. 2019), and to a decline in low-quality housing prices, decreasing the wealth of the less-well off (Ajzenman, Galiani and Seira 2015). The drug war also imposed other negative economic costs - it reduced manufacturing employment and output (Utar 2025), and decreased labour-force participation (Velásquez 2020). I add to this literature by noting that the drug war has also led to a decrease in social capital, particularly interpersonal trust; this impact is especially concentrated within Indigenous communities.

Section 2 explains the background of the Mexican Drug War and the historical background of the evolution of Indigenous institutions in Mexico. Section 3 suggests potential pathways for how violence impacts social capital. Section 4 discusses the data used in the analysis. Section 5 outlines the methodology. The results are presented in Section 6. Potential mechanisms for the results are explored in Section 7. Section 8 concludes.

2 Background and Context

2.1 Indigenous Institutions

According to the 2020 Population and Housing Census, 23.2 million people, or approximately 19.4% of the population of Mexico identified as Indigenous. Indigenous communities living in Mexico are diverse, with 68 different Indigenous languages spoken (INEGI 2022). Consequently, different Indigenous communities would likely have different institutions - many of which were preserved due to the nature of Spanish colonization.

In the 1540s, about 20 years after the arrival of the Spanish in present day Mexico, the Spanish instituted a policy whereby they aimed to resettle Indigenous peoples in villages which were called *Pueblos de Indios* (Assies 2008). In a bid to facilitate Spanish rule over Indigenous populations, the Spanish allowed Indigenous communities to keep some level of autonomy - they gave Indigenous elites noble titles and allowed self-government in villages. In exchange, Indigenous communities were expected to pay tribute (Arteaga 2018). Part of paying tribute included supplying the Spanish with individuals who were forced to work for the colonizers (Assies 2008).

By the year 1800, over 4400 *pueblos* were established, containing 90% of the Indigenous population in the region (De Estrada, Lobato, and Miranda 2005). Given that these villages were created based on the borders of Indigenous polities prior to colonization (Arteaga 2018), and that Indigenous communities retained some control over their internal affairs, many pre-colonial Indigenous institutions persisted throughout the colonial period. There is reason to believe that some elements of Indigenous institutions were also preserved post-independence. The organizational structure of villages was maintained after Mexican independence in 1821, with just under three-quarters of all Mexican municipalities today having a historical connection to at least one *pueblo* (Arteaga 2018). It is therefore possible that Indigenous groups with strong institutions before colonization may have been able to maintain institutional strength during colonization and after the establishment of the Mexican state. This implies

that given the diversity of Indigenous groups throughout pre-colonial Mexico, there will be substantial variation in the level of Indigenous institutions - suggesting that heterogenity can be observed based on the strength of local institutions and local social networks - which may help in assessing the impact of violence.

2.2 The Mexican Drug War

In December 2006, soon after the inauguration of President Felipe Calderón, Mexico announced a "War on Drugs". This involved the militarization of drug enforcement actions, where President Calderón primarily used the military to combat drug cartels. The military directly confronted drug cartels, with many leaders of cartels being arrested or killed. The effect is that there was a fragmentation of cartels such that the number of drug trafficking organizations in Mexico rose from six in 2006 to sixteen in 2011. The increased number of organizations also led to more violence as the divided organizations fought each other to maintain control over their assets and territory. This, combined with directly fighting government forces more often, lead to an increase of violence, such that violence had reached then-record levels by mid 2008. In addition, some of the fighting spread to large swaths of Mexican society, even in municipalities that had not seen high crime levels before. For example, in 2007, only 53 municipalities had at least one death per month on average, while in 2010, that number rose to 200 municipalities (Guerrero-Gutiérrez 2011). Consequently, the increase in violence can be thought of as plausibly exogenous arising from the Mexican government deciding to militarize efforts to combat drug cartels.

3 Relationship to Theory

In this section, I will summarize some theories of how violence impacts social capital. Although there exist many theories, we might have special reason to believe that two might be especially pertinent for the local context in Mexico.

In their review paper, Bauer et al. (2016) briefly outline one theory - lasting gains to social capital when cooperation becomes the optimal choice. Violence leads to the destruction of assets, meaning that investing in physical capital would become less profitable. Therefore, people can benefit from investing time and resources in social capital. This would facilitate participation in informal insurance schemes that may allow individuals to rely on the resources of friends or family when needed. In addition, community participation could allow communities to protect themselves from an outside threat. For instances, investments in social capital to build neighbourhood watch patrols may be helpful in warding off violence or theft. This cooperative behaviour in the face of violence may persist even after the wave of violence subsides. With an entire community behaving in a more pro-social way, individual beliefs about the pro-social behaviour of others may change, driving the community to a higher cooperative equilibrium.

There may be reason to believe that this could be relevant in the context of Mexico. Angeles and Elizalde (2017) argue, Amerindian populations throughout Latin America, but especially in Mexico and Bolivia, organized and attempted to defend their interests against national governments. Communities that were able to do this received more favourable treatment, including better educational and health services, and access to a greater amount of arable land.[2] Consequently, if communities were able to defend their rights from the government, it stands to reason that they might have sought to protect their property and livelihoods in the face of a rise in violence.

A different mechanism could also be important. For example, violence often causes psychological distress and trauma. People may become fearful and avoid people and places that remind them of their trauma. Individuals may become more pessimistic, develop negative sentiments about themselves and others, and have less hope for the future. As a result, they could become less inclined to engage and interact with others (Bauer et al, 2016). This

[2]It was typically communities with strong institutional capacity that were successful in organizing and defending their interests - these were typically communities with connections to strong precolonial Indigenous institutions.

could consequently result in reductions in social capital as some individuals withdraw from interacting with others in the local community.

There may be reason to believe that a post-traumatic stress response based on fear could be important. Brown et al. (2019) use the surge of violence in Mexico due to the militarization of the drug war to estimate the impact of violence on risk aversion. Not only do the authors find that violence increases risk aversion, but they find that feelings of fear and insecurity is likely the underlying mechanism. Consequently, if fear is related to changing risk aversion preferences, then perhaps it may be related to preferences regarding interpersonal trust.

4 Data

Data are drawn from various sources. Data on levels of social trust is taken from the Mexican Family Life Survey, a nationally representative panel-based survey. Data on levels of violence were taken from the National Institute of Statistics and Geography (INEGI), a statistical agency in Mexico. I finally used Murdock's (1967) *Ethnographic Atlas* to construct a measure of Indigenous institutions.

4.1 Mexican Family Life Survey

Firstly, I used data (the second and third waves) from the Mexican Family Life Survey (MxFLS).[3] The MxFLS is a nationally representative panel survey with a 2002 baseline, surveying 35,000 individuals from 8400 households across 150 urban and rural municipalities throughout the country. The second wave (MxFLS2) was conducted between 2005-2006, while the third wave (MxFLS3) occurred during 2009-2012. Roughly 90% of baseline survey respondents were recontacted in both the second and third wave, suggesting there is not substantial attrition between the waves (Rubalcava and Teruel 2013b). The timing of the

[3]The second and third waves are sourced from (Rubalcava and Teruel 2008) and (Rubalcava and Teruel 2013a) respectively

waves is crucial for the paper. Given that the second wave was conducted while violence was relatively stable and the third wave was conducted after the surge in violence began, data from the second and third waves mesh well with the proposed exogenous increase of violence: the militarization of the Mexican Drug War.[4]

The MxFLS collects various demographic and socioeconomic characteristics of the sample population across the three waves, including both on a household level and on an individual level. Most household level questions were asked to the household head (or another adult if the head was not present at the time of the interview).

Crucial for this paper, the household component of the survey contains measures related to social capital. Most importantly, it asks about community level trust, asking households to Completely Agree, Agree, Disagree, or Completely Disagree with the following statement: *People from this locality/community are trustworthy.* This is the main dependent variable of interest that will be used throughout the paper, being used as a measure of interpersonal trust.

I convert the trust responses into binary variables. I create the variable *Least Trust Level* which indicates if a household completely disagrees with the statement "*People from this locality/community are trustworthy.*" I also create *Disagree Community Trustworthy* which indicates if a household disagrees or completely disagrees with the statement.[5]

In addition to these measures, the MxFLS also contains data on indigeneity. Specifically, individuals are asked if they identify as part of an Indigenous group and if they speak an Indigenous language (but not what language is spoken). For the purposes of this paper, I consider a household Indigenous if the household head or the spouse of the household head

[4]In Figure A1, I construct a measure of homicides between 2001 and 2014, which is a population weighted average of the homicide rate across 2106 Mexican municipalities, by average homicide rate (1 per 10000 people) per year. As shown in the Figure, the homicide rate is relatively stable between 2001-2008, before the surge in violence but then starts to climb in mid 2008.

[5]Other related measures of social capital are also measured. The survey asks households if they Completely Agree, Agree, Disagree, or Completely Disagree with the following questions: *People around here are willing to help their neighbors*, which measures perceptions of how helpful neighbours are and *This neighborhood/community is united and really close*, which measures whether a community is tightly knit and unified. I create similar binary measures for these two variables as well, in a similar fashion to the trust measure.

identifies as Indigenous. Given that I compare households in the second and third waves, I consider households to be Indigenous only if they keep Indigenous status in both waves.

The MxFLS also contains socioeconomic status and democraphic variables including for individuals and households that I use as controls. Selected variables include household head gender, age, and employment, educational status of household head/their spouse, relationship status of household head, household size, whether a household has a telephone, whether it has a cooking room, if it uses gas for cooking, if it has a toilet, and if individuals sleep in the kitchen.[6]

Finally, the MxFLS also contains other variables, such as individual exposure to crime. The MxFLS also contains other measures which I expand upon in the mechanisms section, including a measure of whether individuals feel safe in their homes and if communities organize for self-protection.

4.2 Ethnographic Atlas

Following similar strategies as Gennaioli and Rainer (2007), Michalopoulos and Papaioannou (2013), Angeles and Elizalde (2017), and Elizalde (2020), I use a measure of political complexity of ethnic groups as described in Murdock's (1967) *Ethnographic Atlas* to proxy for strength of pre-colonial Indigenous institutions. The *Ethnographic Atlas* codes various socioeconomic and cultural traits of over 1200 ethnic groups around the globe. In particular, it contains the measure "Jurisdictional Hierarchy beyond the local community level", which takes values 0 to 4, representing organization beyond the community level. 0 represents groups organized in bands, a 1 represents chiefdoms in a single city-state or small villages. A 2 represents larger chiefdoms with many cities, a 3 or 4 represents smaller and larger states respectively, typically with several levels of government. The *Atlas* only records ethnic groups that resided in present-day Mexico to have a value of 0, 1 or 2. In order to construct a measure of Indigenous institutions in each municipality, I follow the procedure

[6]See Appendix Section A1 for more detail.

in Elizalde (2020), and construct a weighted average of Indigenous institutions, using the population of each Indigenous ethnic group in a municipality, based on 1990 INEGI data (INEGI 1990).[7] The weighted average takes the following form:

$$IndigInst_{mj} = \sum_{e} \frac{IndigenousPopulation_{emj} * JH_e}{IndigenousPopulation_{mj}} \qquad \text{(Adapted from Elizalde (2020))}$$

$IndigInst_{mj}$ is a measure of precolonial Indigenous institutions in municipality m in state j. $IndigenousPopulation_{mj}$ is the total population of Indigenous individuals in municipality m, in state j in the year 1990. $IndigenousPopulation_{emj}$ is the total population of Indigenous individuals part of ethnic group e, in municipality m, in state j in the year 1990. JH_e is a measure of political complexity of ethnic group e, based on the *Jurisdictional Hierarchy* measure of the *Ethnographic Atlas*.

I merge this data with the existing MxFLS data to get the average level of Indigenous institutions for each municipality. Given that the MxFLS does not contain questions asking which Indigenous group respondents belong to, I am unable to assign Indigenous households to a specific ethnic group, necessitating the weighted average approach.

The level of Indigenous institutions will enable the observation of any heterogeneous effects of violence based on the level of institutions.

4.3 INEGI

Next, I use INEGI Registered Death Statistics to get a measure of crime trends (INEGI 2025). Included in death statistics are all officially reported homicides in a particular municipality in a particular month. Using this data, I calculate the homicide rate (per 10,000 people) in the past 12 months for each municipality at any given time period, and matched this with the MxFLS data by municipality and survey month, to get the main explanatory variable -

[7] I am grateful to Aldo Elizalde for his help in the construction of this metric.

the homicide rate.

In the analysis to follow, I ensure that all households are measured in the second (2005-2006) and third (2009-2012) waves. I also drop observations with missing values for one of the six social capital dependent variables,[8] and selected socioeconomic and demographic controls.[9] I am left with 6,353 households that have observations in both waves, of which 562 are indigenous.

4.4 Summary Statistics

I report the summary statistics of the six dependent variables (Least Trust Level, Least Help Level, Least Close Level, Disagree Community Trustworthy, Disagree Neighbours Help, and Disagree Community Close.) I compare averages for all 6353 households and the 562 Indigenous Households in the 2nd and 3rd wave of the Mexican Family Life Survey. The results are given in Table 1.

Overall, it seems that in wave 2, before the increase in violence, social capital seemed to be higher compared to in wave 3, both overall and among Indigenous households. It seems that households are more likely to Completely Disagree or Disagree that they can trust members in their community, that their community is close and that people help their neighbours in wave 3 than in wave 2, suggesting that social capital is lower in wave 3.

However, it seems that social capital is higher among Indigenous households than in the full sample in both waves. In both waves, for all dependent variables, the mean is lower for Indigenous households than the full sample[10] - indicating that Indigenous households compared to the overall sample are less likely to completely disagree or disagree that: (a) they can trust members in their community, (b) their community is close, and (c) people help their neighbours.

Finally, it seems that overall, households are unlikely to disagree that their community

[8]These are the binary values related to completely disagreeing with, and completely disagreeing or disagreeing with statements on community trust, closeness, and neighbours' willingness to help.

[9]The selected controls are highlighted in the Appendix.

[10]Note that higher values of the dependent variables imply lower social capital.

is close, trustworthy and individuals help their neighbours. It seems that for both the full sample and the Indigenous subset, in all waves, less than 10 percent completely disagree with any of the three statements. Moreover, less than 30 percent disagree or completely disagree. Consequently, in both waves, over 70 percent agree that their community is close, community members can be trusted, and that people generally help their neighbours - suggesting substantial levels of pre-existing social capital on average.

I report additional socioeconomic summary stats for households in the sample in Table A1.

5 Methodology

The main empirical strategy can be expressed in the following regression equation:

$$Y_{hmt} = \beta_1 Hom_{mt} + \beta_2 Hom_{mt} * Index_m + \theta_h + \gamma_t + \varepsilon_{hmt}, \quad (1)$$

where Y_{hmt} is a dummy outcome variable for a household h in municipality m surveyed in time t. Hom_{mt} is the homicide rate in a particular municipality m at time t per 10,000 people. $Index_m$ represents the weighted average of a measure of precolonial Indigenous institutions - based off Murdock's *Jursidictional Hierarchy Beyond the Local Community* measure. θ_h represents household fixed effects, while γ_t represents time fixed effects, which is composed of wave, year and month fixed effects.[11]

To address potential endogenous sorting due to violence, all households were assigned to the municipality they were located in during the 2nd wave of the survey, which was conducted before the surge in crime.

Using a household fixed effects model is important to address potential individual level heterogeneity that may otherwise lead to correlations between violence and measures of trust. For instance, if more trusting and cooperative individuals are more tolerant of violence

[11]The Mexican Family Life Survey regularly surveyed households over multiple dates. For simplicity, the interview date was taken to be the earliest survey date in each wave.

and willing to reside in high violence areas, a simple OLS regression would overestimate the impact of violence on social trust. The panel-based nature of the MxFLS controls for unobserved individual heterogeneity as it measures changes in the same individuals over time.

The fixed effect model also addresses potential concerns about non random attrition. The fact that not all households in the MxFLS were recontacted in future waves may be concerning if the attrited group is systematically different in their trust related behaviour compared to the remaining sample. However, the utilized model looks at changes in the same households, and thus, results are not driven by attrition.

Therefore, I assert that the fixed effects model will yield the causal effect of an increase of homicide rate on a specific outcome variable. Since the surge in violence during the Mexican Drug War was plausibly exogenous to social capital levels, and since the 2nd wave is conducted before the surge, while the 3rd MxFLS wave occurs after the beginning of the increase in homicides, differences between the 2nd and 3rd wave are likely due to the increase in violence.

In table A2, I verify that there is no correlation between the rise of violence and the level of Indigenous institutions in a municipality for 2087 municipalities across Mexico. (Finding a correlation would suggest that violence is not random.) Using the change in the homicide rate between MxFLS2 and MxFL3 as the explanatory variable, and the level of Indigenous institutions in a municipality as the dependent variable, it seems that upon adding state fixed effects, there is no significant correlation between the rise of violence and the level of Indigenous institutions in a municipality. [12]

[12]The regression model is given by: $JH_{ms} = \Delta MurderRate_{ms} + \delta_s$, where JH_{ms} is the level of Indigenous institutions in a municipality m in state s. $\Delta MurderRate_{ms}$, represents the change in the murder rate in municipality m, in state s, while δ_s represent state fixed effects.

6 Results

I use the above model to estimate how violence impacts social capital, and how this impact varies in municipalities with strong precolonial Indigenous institutions. Table 2 shows how violence impacts community level trust amongst the full sample of households and Indigenous households specifically. The *Murder Rate* gives the impact of violence on trust, while the coefficient for *Murder Rate X JH Index* is an interaction term between the homicide rate and the Indigenous institutions measure, the weighted *Jurisdictional Hierarchy* estimate in each municipality.

Looking first at the full sample of 6,353 households, and looking at the variable Least Trust Level, it appears the point estimate in (1) is not statistically significant. This, however, hides some heterogeneity, as in (2), in municipalities where the measure of Indigenous Institutions is 0, an increase of homicides by 1 per 10000 people increases the proportion of households who completely disagree members in their community can be trust by 0.67 percentage points - and is attenuated by a 0.47 percentage point fall for each unit increase in a municipality's Indigenous institutions measure. However, this latter point estimate is only statistically significant to the 10% level. Looking at the Disagree Trustworthy variable, the point estimate for (3) is not statistically significant, but there seems to be some heterogeneity. While not significant to the 5 percent level, an increase in the homicide rate by 1 murder per 10000 people increases the proportion who distrust their neighbours by 0.95 percentage points for communities with an Indigenous institutions score of 0. For each unit increase in a municipality's Indigenous institutions, the proportion of households who distrust their neighbours fall by 1.19 percentage points (compared to municipalities with non existent Indigenous institutions) on average.

Consequently, it seems that for the overall sample, the surge in violent crime has heterogeneous effects. It reduces social trust, but the reduction of social trust is attenuated and even reversed in municipalities with strong Indigenous institutions.

Looking at the sample of the 562 Indigenous households in Panel B, it seems that the

point estimates increase in magnitude. An increase in the murder rate by 1 per 10,000 individuals results in the proportion of households completely disagreeing that community members are trustworthy to increase by 0.97 percentage points. When adding heterogeneity in specification (2) it appears that in municipalities where the Indigenous institutions measure is 0, violence increases the proportion who are the least trusting of their neighbours by 2.89 percentage points, but this number is attenuated by a 1.73 percentage point fall per unit increase of the Indigenous institutions measure in the municipality.

Overall, in specification (3) an increase in the homicide rate by 1 per 10000 individuals increases the proportion of households who distrust their neighbours by 1.73 percentage points - compared to a mean in the 2nd wave of 16.5%, this is moderately substantial. There is heterogeneity here, as in municipalities with an Indigenous institutions measure of zero, violence increases the proportion of households who distrust their neighbours by 5.44 percentage points, but (though it is only significant to the 10 percent level), this is attenuated by 3.35 percentage points per unit increase of the Indigenous Institutions measure in a municipality. Thus, it seems that violence overall reduces social trust, but the loss to social trust is lessened in municipalities with stronger Indigenous Institutions.

In Table 3, I use an Anderson summary index [13] that is composed of six dependent social capital variables. [14] The index is created such that higher values indicate lower social capital. The mean in the second wave is fixed to 0. Though looking at the total sample in Panel A, it seems the impact of violence has a quite a small effect on social capital, with point estimates being quite small - though there is evidence of attenuation based on institutional strength. An increase in the homicide rate by 1 per 10,000 increases the index measure by 0.031 standard deviations, but this is attenuated by a 0.030 standard deviation drop for each unit increase of the Indigenous institutions measure. Looking at the Indigenous households sample, it appears that an increase in the homicide rate by 1 per 10000 results in an increase

[13] See Anderson (2008).

[14] These include the main two binary trust variables, and analogous variables measuring if a community is tightly knit, and perceptions on how helpful neighbours are.

in the Anderson Index by about 0.051 standard deviations. However, for municipalities with an Indigenous institutions measure of zero, the rise in the Weak Social Capital Index is 0.183 standard deviations, but this is attenuated by a 0.119 standard deviation drop for each unit increase of the Indigenous institutions measure in a municipality.

I also look at the effects for two other measures of social capital - perceptions of whether neighbours are willing to help, and whether the community is united in Tables A3 and A4 respectively. Overall the results from these tables follow similar patterns as Tables 2 and 3. Among the Indigenous household sample, there is a modest fall in social capital variables in response to a rise in the murder rate; this fall is more pronounced in municipalities with weaker Indigenous institutions, but is attenuated, (or in some cases reversed) in areas with stronger Indigenous institutions. Overall, the point estimates for the total sample are smaller than for the Indigenous household subset and though often lacking significance, follow the same general pattern of heterogeneous estimates based on the strength of Indigenous institutions. This suggests that the Indigenous subsample may be driving the results.

There could be omitted variable bias that is captured in the results, particularly in the *Jurisdictional Hierarchy* x Homicide measure result, and in fact something else correlated with Indigenous institutions is impacting the social capital variables. We might expect various socioeconomic controls to be correlated with Indigenous institutions [15]. To address this, I use socioeconomic status and demographic controls and interact them with the homicide rate to see if the result for Indigenous institutions on social trust (in this case whether an individual disagrees or completely disagrees they can trust their community members) can be explained by socioeconomic or democraphic controls in table A5. I also interact the proportion of Indigenous individuals in a municipality with violence to ascertain if the proportion of Indigenous members, rather than the strength of Indigenous institutions is driving the effect. This being a driver would lend support to there being some parochial impact. Overall, it seems the estimates are somewhat robust. Looking at the point estimate for the murder rate

[15] One reason for this might be if for instance areas with higher institutional capacity were able to more successfully extract concessions from the government, see Elizalde (2020).

(or the change in the proportion of households who disagree their neighbours are trustworthy in municipalities with weak Indigenous institutions), for most specifications, the point estimate is within 2 percentage points of the point estimate for (1). The exceptions are when the murder rate is interacted with having a toilet, with a substantially higher point estimate, when the murder rate is interacted with relationship status of the head of household, with a higher point estimate, when Indigenous proportion is being considered (the point estimate is drastically higher with a rise in the murder rate by 1 per 10000 resulting in a point estimate of 13.2 percentage points) and column 15, where all interaction effects. In (15), the point estimate is 27.3 percentage points.

Looking at the point estimate for the interaction term between the Murder Rate X JH Index, it appears the point estimate is roughly stable. The exception is when the murder rate is interacted with having a toilet, indigenous proportion, and when all SES variables are interacted with. Most point estimates are significant to at least the 10 percent level.

Since the point estimates are roughly stable, it appears that the results are generally robust to the SES variables. Consequently, this suggests that accounting for SES controls does not drastically change the results.

Table A6 is similar but uses the the proportion of individuals who strongly disagree that their neighbours are trustworthy as the outcome variable. Similar to the results in table A5, the point estimates are broadly stable for most controls. The most glaring exception is (14), which is where the murder rate and the proportion of Indigenous individuals in a municipality are interacted with, which results in a larger in magnitude point estimate for the murder rate, but a lower in magnitude point estimate for the interaction term between the murder rate and Indigenous institutions.

Note however, that while most point estimates are significant to at least the 10 percent level, the two main point estimates for the final regression specification (15) are not - but this could be due to the amount of interaction terms estimated in the specification.

Overall, it seems that SES and selected demographic controls do not explain the results

for the impact of violence on trust. However, when interacting violence and the proportion of Indigenous individuals in a municipality, the coefficient on the interaction term between *Jurisdictional Hierarchy* and violence falls. This implies that the attentuation from increased Indigenous institutions is reduced after talking into account the proportion of Indigenous population. This suggests some of the attenuation is coming from homogeneity of individuals, meaning that parochialism may be driving some of the results.

7 Mechanisms

Multiple mechanisms have been proposed for how violence affects social capital. In this section, I aim to investigate what mechanism might explain the results I have found. Consequently, in this section I attempt to determine what mechanism might explain the results in Section 6, and specifically whether any mechanism can also explain the heterogeneity in the effect of violence on trust based on the strength of Indigenous institutions.

Community organization in the face of conflict could be a mechanism that affects trust. If communities can organize and protect themselves successfully from violence, it could create trust among community members. Consequently, municipalities with stronger precolonial Indigenous institutions who therefore might have a greater ability to organize may generate higher trust or lower distrust relative to municipalities with weak Indigenous institutions. This could conceivably explain the heterogeneity in Section 6.

I test whether the community organization hypothesis could be a potential mechanism. To do this, I use a specific question in the MxFLS: *Are there organized groups of neighbors who protect or defend themselves?* Households can agree or disagree with the question.

As shown in Table 4, looking at specification (1), an increase in the homicide rate by 1 per 10000 people reduces the proportion of households indicating that there are organized groups of neighbours by 1.62 percentage points. This is statistically significant to the 1 percent level.

In specification (2), there is suggestive evidence of heterogeneity - the point estimates indicate that a rise in the murder rate by 1 per 10000 people reduces the proportion of households who indicate there are organized groups of neighbours by 2.02 percentage points in municipalities with an Indigenous institutions measure of 0, but this drop is attenuated by a 0.36 percentage point rise for each unit increase of the Indigenous institutions measure - however, the estimates for (2) lack statistical significance. Moreover, given that the mean value in the second wave is 14.6%, a point estimate of 0.00361 in the interaction term is almost two orders of magnitude lower than the mean value and, therefore, seems insubstantial.

It is therefore unlikely that the effect of organized communities is driving the effect on social capital; in order to explain the results in Section 6 there would need to be heterogeneous impacts based on Indigenous institutions. The fact there do not seem to be heterogeneous impacts suggests community organization is not a main driver of results.

However, additional supporting evidence might support the hypothesis of organization of communities into groups for protection. With an increase in crime, individuals should be at greater risk of assault, so self reports of assaults should increase compared to before the increase in crime. In neighbourhoods where there are groups organized for protection, this increase in reports should be attenuated. If we suspect that the organized groups mechanism is driving the heterogeneity in results, then it should be the case that municipalities with higher Indigenous institutions have lower increases in reported assaults.

I create a dummy variable for whether a household was assaulted in the last 12 months by using individual level data in the MxFLS. A household is assaulted if an individual in the household was assaulted in the last 12 months. I test the results in Table 5, using the same specifications as in earlier tables, except now whether a household's members were assaulted in the last year is the dependent variable.

None of the results are statistically significant and the point estimate of specification (1) is almost 50 times lower than the mean value in the second wave, suggesting it is also not substantial. There may be heterogeneity. Looking at specification (2), it appears that

though these measures lack significance, an increase of 1 murder per 10000 people results in an increase in the proportion of households with a member assaulted in the last year by 0.40 percentage points, yet this is attenuated by 0.28 percentage points for each level of the Indigenous Institutions measure in a municipality. Given however, that the standard errors are more than the point estimates, these are not precisely estimated, and given that the point estimates are less than 10 percent of the mean of the dependent variable in the 2nd wave, the estimates are not very substantial. This suggests that evidence of heterogeneous effects is weak.

In Table A7, I investigate similar results, but for the proportion of households who know victims of robbery, assault and kidnappings that occurred in the last 12 months - reasoning that more people may be more forthcoming about misfortune befalling others. Overall, results are not statistically significant - standard errors are larger than point estimates, suggesting that there is not a strong effect and the results are weak. Consequently overall, there is not strong evidence to support the case that community organization is the mechanism driving the results for social capital.

Another important potential mechanism is related to theories on post-traumatic withdrawal; violence increases psychological trauma and distress, making individuals withdraw from social interactions, thereby possibly negatively harming social capital.

I test for whether distress and fear might potentially explain the results. In the MxFLS, households were asked to complete the sentence "in this house, do you feel", with the following options: (a) *Very safe*, (b) *Safe*, (c) *Unsafe*, and (d) *Very unsafe*.

I investigate whether violence affects the proportion of people who feel 'unsafe' or 'very unsafe' in Table 6. As shown in the table, overall, an increase in violence by 1 per 10,000 does not seem to have a significant impact on the proportion of people who feel unsafe in their home. However, this masks heterogeneity by the strength of Indigenous institutions in a particular municipality. It appears that for municipalities with an Indigenous institutions measure of 0, an increase in the homicide race increases the proportion of households who

feel unsafe in their homes by 3.20 percentage points - yet this effect is attenuated by about 3.21 percentage points for every unit increase in the Indigenous institutions measure for the municipality of the household. The impact of violence on feeling unsafe is therefore attenuated in municipalities with strong precolonial Indigenous institutions. In particular, it seems that in municipalities with an Indigenous institutions score of one, the effects of fear seem to be completely eliminated.

The fact that these estimates move in the same directions as the estimates for trust and other variables associated with social capital imply that these might be related, and therefore there exists some evidence that feeling unsafe, and therefore fearful, is the channel for which trust is impacted. In particular, feeling unsafe may harm social trust (and therefore social capital more broadly).

I therefore argue it is likely that fear is the mechanism that explains the impact of trust in Mexico - and something about strong Indigenous institutions is able to limit increases in fear, and therefore lead to lower loss in interpersonal trust and social capital.

8 Conclusion

Overall, it seems that the increase in violence caused by the Mexican Drug War has led to losses in social capital and interpersonal trust specifically. This is more pronounced for Indigenous households, compared to a broader sample of households. However, losses to social capital are mitigated (in areas with strong Indigenous institutions that stem from precolonial times. This suggests that something about institutions themselves may matter for social capital in the face of violence. The results are broadly robust to socioeconomic status controls and follow similar patterns for different determinants of social capital (willingness of neighbours to help others, but especially perceptions of community unity and the degree to which households trust community members).

Looking at potential mechanisms, there is evidence to suggest that an increase in violence

decreased the propensity of communities to organize to defend themselves, but not evidence of heterogeneous effects that are crucial to explain the results in this paper. Looking at fear however, it appears that violence increases the degree to which households feel unsafe in their homes - yet this is attenuated for households in municipalities with strong Indigenous institutions. This suggests that fear may be driving the heterogeneous social capital results.

The implication is that fear might be a driving factor in Mexico, with violence reducing social capital due to creating fear - possibly leading individuals to withdraw from social interactions. Interestingly, communities with strong Indigenous institutions are able to avoid this increase in fear and avoid losses due to social capital, suggesting that households in these communities are able to cope with violence more easily. A plausible explanation is that communities with stronger institutions have supports that enable people to feel safe and less afraid - but more work needs to be done to investigate why there is less of an increase in fear in communities with stronger Indigenous institutions.

The fact that losses to social capital in the face of rising crime are lessened with strong Indigenous institutions is interesting - future work should investigate whether this is the effect of *Indigenous* precolonial institutions, or whether similar effects might be found for other local institutions. If in fact results are generalizable to local institutions more broadly, building strong local institutions could serve as ways to preserve social capital during times of unrest or upheaval - facilitating the rebuilding process in communities torn apart by crime or violence.

References

Ajzenman, Nicolas, Sebastian Galiani, and Enrique Seira, "On the Distributive Costs of Drug-Related Homicides," *The Journal of Law and Economics*, November 2015, *58* (4), 779–803.

Anderson, Michael L., "Multiple Inference and Gender Differences in the Effects of Early

Intervention: A Reevaluation of the Abecedarian, Perry Preschool, and Early Training Projects," *Journal of the American Statistical Association*, December 2008, *103* (484), 1481–1495.

Angeles, Luis and Aldo Elizalde, "Pre-colonial institutions and socioeconomic development: The case of Latin America," *Journal of Development Economics*, January 2017, *124*, 22–40.

Arteaga, Fernando, "The Historical Legacy of (Pre?) Colonial Indigenous Settlements in Mexico," 2018.

Assies, Willem, "Land Tenure and Tenure Regimes in Mexico: An Overview," *Journal of Agrarian Change*, January 2008, *8* (1), 33–63.

Bauer, Michal, Christopher Blattman, Julie Chytilová, Joseph Henrich, Edward Miguel, and Tamar Mitts, "Can War Foster Cooperation?," *Journal of Economic Perspectives*, 2016, *30* (3), 249–74.

Becchetti, Leonardo, Pierluigi Conzo, and Alessandro Romeo, "Violence, trust, and trustworthiness: evidence from a Nairobi slum," *Oxford Economic Papers*, January 2014, *66* (1), 283–305.

Bellows, John and Edward Miguel, "War and local collective action in Sierra Leone," *Journal of Public Economics*, December 2009, *93* (11), 1144–1157.

Bogliacino, Francesco, Camilo Gómez, and Gianluca Grimalda, "Crime-related exposure to violence and prosocial behavior: Experimental evidence from Colombia," *Journal of Behavioral and Experimental Economics*, June 2023, *104*, 102023.

Brown, Ryan, Verónica Montalva, Duncan Thomas, and Andrea Velásquez, "Impact of Violent Crime on Risk Aversion: Evidence from the Mexican Drug War," *The Review of Economics and Statistics*, December 2019, *101* (5), 892–904.

de Estrada, Dorothy Tanck, José Antonio Álvarez Lobato, and Jorge Luis Miranda, "Atlas ilustrado de los pueblos de indios de la Nueva España, 1800," *Journal of Latin American Geography*, 2005, *4* (2), 97–109.

Dell, Melissa, Nathan Lane, and Pablo Querubin, "The Historical State, Local Collective Action, and Economic Development in Vietnam," *Econometrica*, November 2018, *86* (6), 2083–2121.

Durlauf, Steven N. and Marcel Fafchamps, "Chapter 26 - Social Capital," in Philippe Aghion and Steven N. Durlauf, eds., *Handbook of Economic Growth*, Vol. 1, Elsevier, January 2005, pp. 1639–1699.

Elizalde, Aldo, "On the economic effects of Indigenous institutions: Evidence from Mexico," *Journal of Development Economics*, November 2020, *147*, 102530.

Fiedler, Charlotte, "What Do We Know about How Armed Conflict Affects Social Cohesion? A Review of the Empirical Literature," *International Studies Review*, September 2023, *25* (3), viad030.

Gennaioli, Nicola and Ilia Rainer, "The modern impact of precolonial centralization in Africa," *Journal of Economic Growth*, September 2007, *12* (3), 185–234.

Guerrero-Gutiérrez, Eduardo, "Security, Drugs, and Violence in Mexico: A Survey," 2011. Paper presented at the 7th North American Forum, Washington, DC.

Hoch, Georg, Andreas Pondorfer, and Viktoriia Shkola, "Conflict and social capital: Evidence from the Russian War against Ukraine," *Journal of Comparative Economics*, June 2025, *53* (2), 461–471.

INEGI, "XI General Census of Population and Housing (CPV) 1990," 1990.

_ , "XII General Census of Population and Housing (CPV) 2000," 2000.

_ , "II Count of Population and Housing (Conteo) 2005," 2005.

_ , "Census of Population and Housing (CPV) 2010," 2010.

_ , "Sala de prensa," August 2022.

_ , "Registered Death Statistics," 2025.

Kijewski, Sara and Markus Freitag, "Civil War and the Formation of Social Trust in Kosovo: Posttraumatic Growth or War-related Distress?," *Journal of Conflict Resolution*, April 2018, *62* (4), 717–742.

Knack, Stephen and Philip Keefer, "Does Social Capital Have an Economic Payoff? A Cross-Country Investigation," *The Quarterly Journal of Economics*, 1997, *112* (4), 1251–1288.

Michalopoulos, Stelios and Elias Papaioannou, "Pre-Colonial Ethnic Institutions and Contemporary African Development," *Econometrica*, January 2013, *81* (1), 113–152.

Murdock, George Peter, "Ethnographic Atlas: A Summary," *Ethnology*, 1967, *6* (2), 109–236.

Rohner, Dominic, Mathias Thoenig, and Fabrizio Zilibotti, "Seeds of distrust: conflict in Uganda," *Journal of Economic Growth*, September 2013, *18* (3), 217–252.

Rubalcava, Luis and Graciela Teruel, "Mexican Family Life Survey, Second Wave," 2008.

_ **and** _ , "Mexican Family Life Survey, Third Wave," 2013a.

_ **and** _ , "User's Guide for the Mexican Family Life Survey Third Wave," 2013b.

Rubio-Ramos, Melissa, "Trust, violence, and coca," *Journal of Development Economics*, March 2024, *167*, 103216.

Utar, Hale, "Firms and Labor in Times of Violence: Evidence from the Mexican Drug War," *The World Bank Economic Review*, August 2025, *39* (3), 632–662.

Velásquez, Andrea, "The Economic Burden of Crime," *Journal of Human Resources*, October 2020, *55* (4), 1287.

Voors, Maarten J., Eleonora E. M. Nillesen, Philip Verwimp, Erwin H. Bulte, Robert Lensink, and Daan P. Van Soest, "Violent Conflict and Behavior: A Field Experiment in Burundi," *American Economic Review*, 2012, *102* (2), 941–64.

Tables

Table 1: Summary Statistics for Social Capital Variables Among All Households and Indigenous Households in Wave 2 and 3 of the MxFLS.

Panel A: Total Sample

	Wave 2		Wave 3	
	Households	Mean	Households	Mean
Least Trust Level	6,353	0.0261	6,353	0.0535
Least Help Level	6,353	0.0315	6,353	0.0721
Least Close Level	6,353	0.0411	6,353	0.0688
Disagree Trustworthy	6,353	0.219	6,353	0.225
Disagree Neighbours Help	6,353	0.268	6,353	0.275
Disagree Community Close	6,353	0.331	6,353	0.314

Panel B: Indigenous Households

	Wave 2		Wave 3	
	Households	Mean	Households	Mean
Least Trust Level	562	0.0196	562	0.0445
Least Help Level	562	0.0178	562	0.0712
Least Close Level	562	0.0302	562	0.0427
Disagree Trustworthy	562	0.165	562	0.198
Disagree Neighbours Help	562	0.210	562	0.240
Disagree Community Close	562	0.272	562	0.237

Notes: This table shows summary statistics of social capital variables, documenting the proportion of households disagreeing with social capital statements, in the 2nd (2005-2006) and 3rd (2009-2012) of the Mexican Family Life Survey. Results for the total sample of households and Indigenous households are shown separately. *Least Trust Level* indicates if a household completely disagrees with the statement *People from this locality/community are trustworthy*; *Disagree Trustworthy* indicates if a household disagrees or completely disagrees with the statement. *Least Help Level* indicates completely disagreeing with *People around here are willing to help their neighbors*; *Disagree Neighbours Help* indicates disagreeing or completely disagreeing. *Least Close Level* indicates completely disagreeing with *This neighborhood/community is united and really close*; *Disagree Community Close* is an indicator for disagreeing or completely disagreeing.

Table 2: Impact of Violence on Community Level Trust

Panel A: Total Sample

	(1) Least Trust Level	(2)	(3) Disagree Trustworthy	(4)
Murder Rate	0.00376	0.00668**	0.00221	0.00954*
	[0.00234]	[0.00321]	[0.00479]	[0.00564]
Murder Rate X JH Index		-0.00473*		-0.01187**
		[0.00271]		[0.00523]
Observations	12,706	12,706	12,706	12,706
Number of Households	6,353	6,353	6,353	6,353
Year Month Wave FE	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Sample	All	All	All	All
Mean Dep. Variable (2nd Wave)	0.0261	0.0261	0.219	0.219
Murder Rate + JH Interaction		0.00195		-0.00234
p-value		0.295		0.604

Panel B: Indigenous Households

	(1) Least Trust Level	(2)	(3) Disagree Trustworthy	(4)
Murder Rate	0.00965***	0.02887***	0.01734***	0.05444**
	[0.00305]	[0.00912]	[0.00322]	[0.02274]
Murder Rate X JH Index		-0.01733**		-0.03346*
		[0.00848]		[0.01996]
Observations	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Sample	Indigenous	Indigenous	Indigenous	Indigenous
Mean Dep. Variable (2nd Wave)	0.0196	0.0196	0.165	0.165
Murder Rate + JH Interaction		0.0115		0.0210
p-value		1.06e-06		0.000392

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. *Least Trust Level* indicates if a household completely disagrees with the statement *People from this locality/community are trustworthy*; *Disagree Trustworthy* indicates if a household disagrees or completely disagrees with the statement. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects. Panel A presents the results for the total sample, and Panel B presents the results for Indigenous households.

Table 3: Impact of Crime on Social Capital

Panel A: Total Sample

	Anderson Index Weak Social Capital	
Murder Rate	0.01253	0.03076*
	[0.01361]	[0.01683]
Murder Rate X JH Index		-0.02954*
		[0.01576]
Observations	12,706	12,706
Number of Households	6,353	6,353
Year Month Wave FE	Yes	Yes
Household FE	Yes	Yes
Sample	All	All
Mean Dep. Variable (2nd Wave)	0	0
Murder Rate + JH Interaction		0.00123
p-value		0.923

Panel B: Indigenous Households

	Anderson Index Weak Social Capital	
Murder Rate	0.05113***	0.18271***
	[0.01545]	[0.04742]
Murder Rate X JH Index		-0.11867***
		[0.04085]
Observations	1,124	1,124
Number of Households	562	562
Year Month Wave FE	Yes	Yes
Household FE	Yes	Yes
Sample	Indigenous	Indigenous
Mean Dep. Variable (2nd Wave)	0	0
Murder Rate + JH Interaction		0.0640
p-value		5.49e-07

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. *Anderson Index Weak Social Capital* is an Anderson (2008) summary index composed of different social capital indicators relating to trust, how tightly knit a community is, and perceptions as to whether community members help their neighbours. The index is constructed so that higher values indicate lower social capital. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects. Panel A presents the results for the total sample, and Panel B presents the results for Indigenous households.

Table 4: Impact of Violence on Community Organization for Protection.

	Organized Communities	
Murder Rate	-0.01620***	-0.02018
	[0.00445]	[0.01934]
Murder Rate X JH Index		0.00361
		[0.01881]
Observations	1,110	1,110
Number of Households	555	555
Year Month Wave FE	Yes	Yes
Household FE	Yes	Yes
Sample	Indigenous	Indigenous
Mean Dep. Variable (2nd Wave)	0.146	0.146
Murder Rate + Interaction		-0.0166
p-value		0.000409

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Organized Communities* indicates whether households agree with the question *Are there organized groups of neighbors to protect or defend themselves?* Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects.

Table 5: Impact of Violence on Being Assaulted Among Indigenous Households.

	Assaulted Last 12 Months	
Murder Rate	0.00091	0.00399
	[0.00136]	[0.01139]
Murder Rate X JH Index		-0.00277
		[0.00959]
Observations	1,124	1,124
Number of Households	562	562
Year Month Wave FE	Yes	Yes
Household FE	Yes	Yes
Sample	Indigenous	Indigenous
Mean Dep. Variable (2nd wave)	0.0427	0.0427
Murder Rate + JH Interaction		0.00121
p-value		0.565

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. *Assaulted Last 12 Months* is an indicator for if any individual in a household was assaulted in the last 12 months. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects.

Table 6: Impact of a Rise in the Murder Rate on Feeling Unsafe in the Home Among Indigenous Households.

	Unsafe Home	
Murder Rate	-0.00353	0.03201**
	[0.00644]	[0.01400]
Murder Rate X JH Index		-0.03205**
		[0.01268]
Observations	1,124	1,124
Number of Households	562	562
Year Month Wave FE	Yes	Yes
Household FE	Yes	Yes
Sample	Indigenous	Indigenous
Mean Dep. Variable (2nd Wave)	0.0658	0.0658
Murder Rate + JH Interaction		-4.16e-05
p-value		0.991

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. *Unsafe Home* is an indicator for if the household respondent indicates they feel unsafe or very unsafe in their home. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects.

Appendix

A1. Socioeconomic Status and Demographic Variables

As mentioned in the Data section, the MxFLS contains socioeconomic status (SES) and demographic information. In this paper, I dropped households with missing observations for selected SES and demographic variables. The variables are listed as follows:

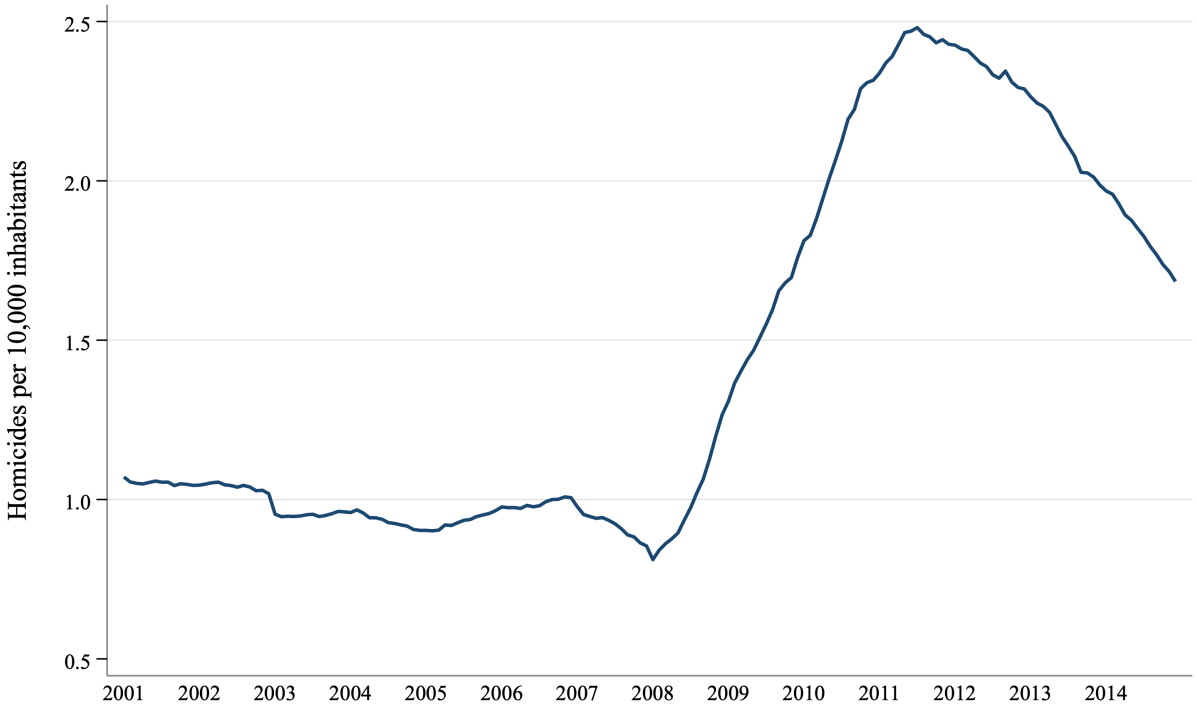
- Has Telephone: A dummy variable indicating if a household has a telephone.
- Has Cooking Room: A dummy variable indicating if a household has a cooking room.
- Sleeps in Kitchen: A dummy indicating if household members sleep in the kitchen.
- Has Toilet: Indicating if a toilet is located inside a household.
- Uses Gas for Cooking: Indicating if a household uses gas for heating while cooking.
- Head Not Single: Indicating if the household head is in a relationship or not.
- Household Size: A variable indicating the number of individuals in the household.
- Age of Head: A discrete variable providing the age of the household head.
- Head Male: A dummy variable that takes a value of 1 if the head of the household is male and 0 if they are female.
- Head or Spouse Employed: A variable indicating if either the head of household or their spouse is employed.
- Head or Spouse Attended Jr. High: A variable indicating if either the head of household or their spouse attended Jr. High.
- Head or Spouse Attended College: A variable indicating if either the head of household or their spouse attended college.

Note that to avoid potential endogeneity in the controls, I fix the values to be the values from the 2nd wave of the MxFLS, before the surge in violence.

These variables, along with the proportion of Indigenous individuals in a municipality, are used as controls in Tables A5 and A6.

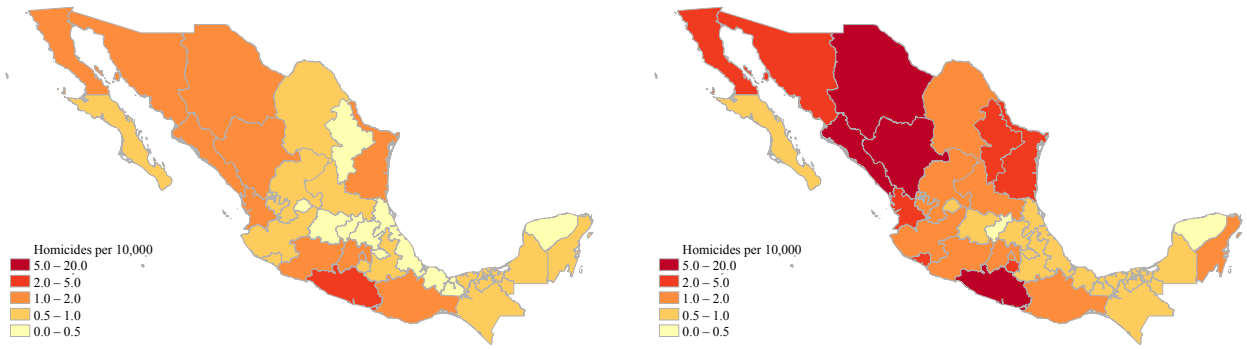
Appendix Figures and Tables

Figure A1: Overall Trend of Homicides Between 2001 and 2014 for 2106 municipalities in Mexico.



Notes: Figure A1 is the weighted average of the homicide rate for 2106 Mexican municipalities between 2001-2014. The homicide rate measures the number of homicides in the last 12 months in a municipality per 10,000 people. The weighted average is calculated by weighting the homicide rate by the population in each municipality. For the years 2001 and 2002, population counts are taken from INEGI (2000). For 2003-2007, population counts are taken from INEGI (2005). For 2008-2014 population counts are taken from INEGI (2010).

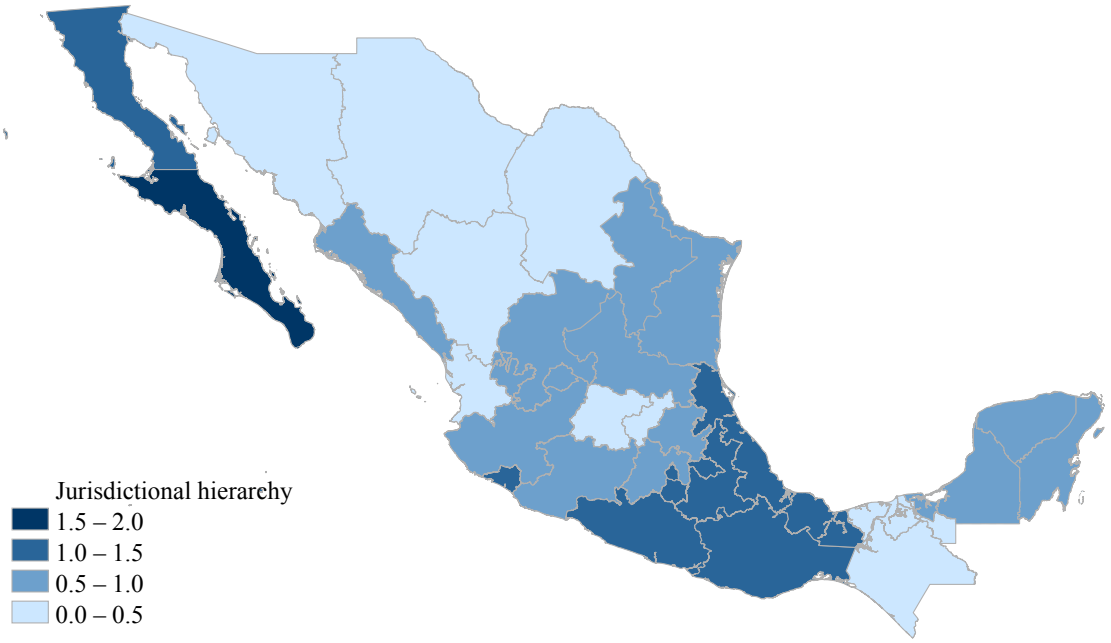
Figure A2: Spatial Distribution of Homicide Rates Across Mexican Municipalities in MxFLS Waves 2 and 3.



(a) Wave 2: 2005–2006 (b) Wave 3: 2009–2012

Notes: These figures map the homicide rate (homicides per 10,000 people in the past 12 months). Municipality-level homicide rates are collapsed to state-level means, weighted by municipality population (2005 levels for Wave 2 and 2010 levels for Wave 3). Wave 2 corresponds to survey years 2005–2006 and Wave 3 corresponds to survey years 2009–2012.

Figure A3: Spatial Distribution of Indigenous Institutions Across Mexican States.



Notes: This figure maps the Indigenous institutions measure used in the analysis, based on the *Jurisdictional Hierarchy Beyond the Local Community* variable from Murdock's (1967) *Ethnographic Atlas*. State-level means are constructed by taking the average across municipalities, weighted by municipality population in 2000.

Table A1: Summary Statistics for Socioeconomic Status Variables Among Households in Wave 2 of the MxFLS.

	Full Sample		Indigenous Households	
	Households	Mean	Households	Mean
Has Telephone	6,353	0.459	562	0.239
Has Cooking Room	6,353	0.928	562	0.879
Sleeps in Kitchen	6,353	0.0964	562	0.148
Has Toilet	6,353	0.754	562	0.407
Uses Gas for Cooking	6,353	0.850	562	0.420
Head Not Single	6,353	0.770	562	0.847
Household Size	6,353	4.736	562	5.464
Age of Head	6,353	49.15	562	51.25
Head Male	6,353	0.796	562	0.843
Head or Spouse Employed	6,353	0.808	562	0.835
Head or Spouse Attended Jr. High	6,353	0.435	562	0.221
Head or Spouse Attended College	6,353	0.0937	562	0.0374

Notes: This table provides the means in wave 2 of the Mexican Family Life Study for demographic and socioeconomic status variables, both for the full sample, and the Indigenous household sub sample. Details on the variables are in Section A1.

Table A2: Correlation Between Rise of Homicides and Indigenous Institutions.

	(1) JH Index	(2) JH Index
Change in Murder Rate	-0.03065***	-0.00224
	[0.00767]	[0.00388]
Municipalities	2,087	2,087
State FE	No	Yes
Average Jurisdictional Hierarchy Value	1.179	1.179

Notes: Change in Murder Rate captures the difference in the homicide rate between years in wave 2 of the MxFLS (2005, 2006) and wave 3 (2009, 2010, 2011, 2012). *JH Index* represents the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. Standard robust errors are reported in parentheses; *10%, **5%,***1% significance levels.

Table A3: Impact of Crime on Perception of Willingness of Neighbours to Help

Panel A: Total Sample

	(1)	(2)	(3)	(4)
	Least Help Level		Disagree Neighbours Help	
Murder Rate	0.00005	0.00109	-0.00098	0.00336
	[0.00211]	[0.00287]	[0.00497]	[0.00622]
Murder Rate X JH Index		-0.00169		-0.00704
		[0.00287]		[0.00652]
Observations	12,706	12,706	12,706	12,706
Number of Households	6,353	6,353	6,353	6,353
Year Month Wave FE	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Sample	All	All	All	All
Mean Dep. Variable (2nd Wave)	0.0315	0.0315	0.268	0.268
Murder Rate + JH Interaction		-0.000600		-0.00368
p-value		0.785		0.500

Panel B: Indigenous Households

	(1)	(2)	(3)	(4)
	Least Help Level		Disagree Neighbours Help	
Murder Rate	0.01126***	0.01871*	0.01993***	0.06749***
	[0.00216]	[0.01045]	[0.00558]	[0.01738]
Murder Rate X JH Index		-0.00672		-0.04289***
		[0.00993]		[0.01516]
Observations	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Sample	Indigenous	Indigenous	Indigenous	Indigenous
Mean Dep. Variable (2nd Wave)	0.0178	0.0178	0.210	0.210
Murder Rate + JH Interaction		0.0120		0.0246
p-value		1.22e-09		1.53e-09

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. *Least Help Level* indicates if a household completely disagrees with the statement *People around here are willing to help their neighbors*; *Disagree Neighbourhood Help* indicates if a household disagrees or completely disagrees with the statement. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects. Panel A presents the results for the total sample, and Panel B presents the results for Indigenous households.

Table A4: Impact of Crime on Community Closeness

Panel A: Total Sample

	(1)	(2)	(3)	(4)
	Least Closeness Level		Disagree Neighbourhood Close	
Murder Rate	0.00399**	0.00606**	0.00263	0.00718
	[0.00202]	[0.00282]	[0.00452]	[0.00618]
Murder Rate X JH Index		-0.00335		-0.00737
		[0.00265]		[0.00589]
Observations	12,706	12,706	12,706	12,706
Number of Households	6,353	6,353	6,353	6,353
Year Month Wave FE	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Mean Dep. Variable (2nd Wave)	0.0411	0.0411	0.331	0.331
Murder Rate + JH Interaction		0.00271		-0.000190
p-value		0.135		0.967

Panel B: Indigenous Households

	(1)	(2)	(3)	(4)
	Least Closeness Level		Disagree Neighbourhood Close	
Murder Rate	0.00537**	0.00645	0.00416	0.06656***
	[0.00251]	[0.01423]	[0.00700]	[0.01884]
Murder Rate X JH Index		-0.00097		-0.05627***
		[0.01188]		[0.01572]
Observations	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes
Mean Dep. Variable (2nd Wave)	0.0302	0.0302	0.272	0.272
Murder Rate + JH Interaction		0.00548		0.0103
p-value		0.0899		0.0296

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. *Least Closeness Level* indicates if a household completely disagrees with the statement *This neighborhood/community is united and really close*; *Disagree Neighbourhood Close* indicates if a household disagrees or completely disagrees with the statement. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects. Panel A presents the results for the total sample, and Panel B presents the results for Indigenous households.

Table A5: Impact of Crime on Proportion of Households Who Disagree or Completely Disagree Their Neighbours can be Trusted, With SES Controls (Part 1 of 2).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Murder Rate	0.05444**	0.05383**	0.03836*	0.06103**	0.08729***	0.05726**	0.05513**	0.07881*
	[0.02274]	[0.02240]	[0.02096]	[0.02413]	[0.02250]	[0.02464]	[0.02293]	[0.04193]
Murder Rate X JH Index	-0.03346*	-0.03266*	-0.03570*	-0.03515*	-0.04828**	-0.03414*	-0.03372*	-0.03573*
	[0.01996]	[0.01946]	[0.01978]	[0.01964]	[0.01910]	[0.01997]	[0.02001]	[0.01981]
Murder Rate X Telephone		-0.01245						
		[0.05027]						
Murder Rate X Cooking Room			0.02313***					
			[0.00801]					
Murder Rate X Sleeps Kitchen				-0.01800				
				[0.01777]				
Murder Rate X Toilet					-0.03285***			
					[0.00377]			
Murder Rate X Jr. High						-0.00880		
						[0.01706]		
Murder Rate X College							-0.10107	
							[0.15483]	
Murder Rate X Head Not Single								-0.02394
								[0.03020]
Observations	1,124	1,124	1,124	1,124	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Dep. Variable (2nd wave)	0.165	0.165	0.165	0.165	0.165	0.165	0.165	0.165

Table A5: Impact of Crime on Proportion of Households Who Disagree or Completely Disagree Their Neighbours can be Trusted, With SES Controls (Part 2 of 2).

	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Murder Rate	0.07040**	0.05929**	0.08231**	0.06468*	0.06415***	0.13228***	0.27365***
	[0.03192]	[0.02334]	[0.03860]	[0.03688]	[0.02306]	[0.04353]	[0.05533]
Murder Rate X JH Index	-0.03371*	-0.03319	-0.03669*	-0.03398*	-0.03662*	-0.02188	-0.05022***
	[0.01987]	[0.02017]	[0.02025]	[0.02000]	[0.02056]	[0.01778]	[0.01861]
Murder Rate X Telephone							-0.00843
							[0.06004]
Murder Rate X Cooking Room							0.02526**
							[0.01081]
Murder Rate X Sleeps Kitchen							-0.04619
							[0.03170]
Murder Rate X Toilet							-0.03335***
							[0.00991]
Murder Rate X Jr. High							-0.00241
							[0.04939]
Murder Rate X College							-0.09661
							[0.15470]
Murder Rate X Head Not Single							-0.04529**
							[0.01909]
Murder Rate X HH Size	-0.00319						-0.00795
	[0.00411]						[0.00630]
Murder Rate X Age Head		-0.00009					-0.00066
		[0.00032]					[0.00060]
Murder Rate X Head is Male			-0.02688				0.00679
			[0.02325]				[0.02569]
Murder Rate X Head Employed				-0.01113			0.00580
				[0.02586]			[0.02463]
Murder Rate X Gas Used					-0.02342		-0.02353
					[0.01455]		[0.03485]
Murder Rate X Indigenous Prop						-0.10796**	-0.09293
						[0.05152]	[0.06302]
Observations	1,124	1,124	1,124	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Dep. Variable (2nd wave)	0.165	0.165	0.165	0.165	0.165	0.165	0.165

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. Each column from (2) to (13) interacts the murder rate with a demographic or socioeconomic status variable. (These variables are further described in Section A1.) Column (14) interacts the murder rate with the proportion of Indigenous individuals in a municipality. Column (15) adds all the interactions to the regression. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects.

Table A6: Impact of Crime on Proportion of Households Who Strongly Disagree Their Neighbours can be Trusted, With SES Controls (Part 1 of 2).

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
Murder Rate	0.02887***	0.02856***	0.02326**	0.03161***	0.04732***	0.03333***	0.02887***	0.02616**
	[0.00912]	[0.00955]	[0.01020]	[0.00904]	[0.01019]	[0.00907]	[0.00911]	[0.01229]
Murder Rate X JH Index	-0.01733**	-0.01693*	-0.01811**	-0.01803**	-0.02565***	-0.01841**	-0.01733**	-0.01708*
	[0.00848]	[0.00904]	[0.00865]	[0.00858]	[0.00915]	[0.00850]	[0.00847]	[0.00868]
Murder Rate X Telephone		-0.00635						
		[0.02724]						
Murder Rate X Cooking Room			0.00807**					
			[0.00320]					
Murder Rate X Sleeps Kitchen				-0.00747				
				[0.00469]				
Murder Rate X Has Toilet					-0.01844***			
					[0.00545]			
Murder Rate X Jr. High						-0.01393***		
						[0.00372]		
Murder Rate X College							0.00007	
							[0.04500]	
Murder Rate X Head not Single								0.00266
								[0.00510]
Observations	1,124	1,124	1,124	1,124	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Dep. Variable (2nd Wave)	0.0196	0.0196	0.0196	0.0196	0.0196	0.0196	0.0196	0.0196

Table A6: Impact of Crime on Proportion of Households Who Strongly Disagree Their Neighbours can be Trusted, With SES Controls (Part 2 of 2).

	(9)	(10)	(11)	(12)	(13)	(14)	(15)
Murder Rate	0.01190	0.02620*	0.02701*	0.02251**	0.03345***	0.06197*	0.07125
	[0.01174]	[0.01494]	[0.01378]	[0.01116]	[0.00837]	[0.03608]	[0.04705]
Murder Rate X JH Index	-0.01706*	-0.01748**	-0.01712*	-0.01701*	-0.01882**	-0.01241	-0.02064
	[0.00860]	[0.00830]	[0.00874]	[0.00855]	[0.00811]	[0.00942]	[0.01343]
Murder Rate X Telephone							-0.02557
							[0.02833]
Murder Rate X Cooking Room							-0.00121
							[0.00592]
Murder Rate X Sleeps Kitchen							0.00267
							[0.01291]
Murder Rate X Has Toilet							-0.01847
							[0.01305]
Murder Rate X Jr. High							0.00256
							[0.02542]
Murder Rate X College							0.02200
							[0.04226]
Murder Rate X Head not Single							-0.01367
							[0.02178]
Murder Rate X HH Size	0.00339***						0.00488*
	[0.00085]						[0.00245]
Murder Rate X Age Head		0.00005					0.00012
		[0.00016]					[0.00059]
Murder Rate X Head Male			0.00180				-0.00904
			[0.00786]				[0.01149]
Murder Rate X Head Employed				0.00691*			0.00722
				[0.00401]			[0.01517]
Murder Rate X Gas Used					-0.01104**		-0.00619
					[0.00489]		[0.00658]
Murder Rate X Indigenous Prop						-0.04591	-0.05236
						[0.04269]	[0.04435]
Observations	1,124	1,124	1,124	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Mean Dep. Variable (2nd Wave)	0.0196	0.0196	0.0196	0.0196	0.0196	0.0196	0.0196

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. Each column from (2) to (13) interacts the murder rate with a demographic or socioeconomic status variable. (These variables are further described in Section A1.) Column (14) interacts the murder rate with the proportion of Indigenous individuals in a municipality. Column (15) adds all the interactions to the regression. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects.

Table A7: Effect of Violence on Knowing People Victimized Among Indigenous Households.

	(1)	(2)	(3)	(4)	(5)	(6)
	Know Person Robbed		Know Person Attacked		Know Person Kidnapped	
Murder Rate	0.00151	-0.00243	0.00070	0.00360	0.00060	0.00109
	[0.00158]	[0.00813]	[0.00117]	[0.00537]	[0.00065]	[0.00175]
Murder Rate X JH Index		0.00355		-0.00262		-0.00045
		[0.00741]		[0.00499]		[0.00120]
Observations	1,124	1,124	1,124	1,124	1,124	1,124
Number of Households	562	562	562	562	562	562
Year Month Wave FE	Yes	Yes	Yes	Yes	Yes	Yes
Household FE	Yes	Yes	Yes	Yes	Yes	Yes
Sample	Indigenous	Indigenous	Indigenous	Indigenous	Indigenous	Indigenous
Mean Dep. Variable (2nd Wave)	0.0178	0.0178	0.0142	0.0142	0	0
Murder Rate + JH Interaction		0.00112		0.000982		0.000646
p-value		0.504		0.369		0.386

Notes: Murder Rate represents the number of homicides per 10,000 within a municipality in the last 12 months. Murder Rate X JH Index is the homicide rate multiplied by the weighted average of the *Jurisdiction Hierarchy beyond the local community* measure for the municipality. *Know Person Robbed* indicates if a household respondent knows someone who was robbed at home or their place of business within the last 12 months. *Know Person Attacked* indicates if a respondents knows someone assaulted within the last 12 months, while *Know Person Kidnapped* indicates if the respondent knows someone who was kidnapped in the last 12 months. Standard errors are clustered at the municipality level and reported in parentheses; *10%, **5%,***1% significance levels. All specifications use household, year, month and MxFLS Wave fixed effects.